A SECURE PARAMETRIC NETWORK PROTOCOL WITH ROTATING ENCODING MAPS FOR FORWARD SECRECY

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Title: A SECURE PARAMETRIC NETWORK PROTOCOL WITH ROTATING ENCODING MAPS FOR FORWARD SECRECY

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the U.S. Patent Applications for "A Protocol Stack and System for Parametric Data Networking" and "Protocol for Encoding Map Rotation for Forward Secrecy," the contents of which are hereby incorporated by reference in their entirety.

Abstract

A secure network protocol that provides forward secrecy at the data encoding layer within a routed, multi-hop parametric network. The protocol governs communication between two endpoints through one or more intermediate routers. The endpoints use a series of disposable, time-variant encoding maps to encode and decode the payloads of parametric packets. The

intermediate routers, which forward packets based on symbolic headers, have no knowledge of the encoding maps being used. The endpoints periodically rotate to a new map based on a shared, pre-defined trigger. A compromise of an intermediate router or the endpoints' map-generation secret at a given time does not allow for the decryption of previously captured traffic that was encoded with a since-discarded map.

Background of the Invention

The invention of parametric networking introduced a paradigm where data is transported as unevaluated symbolic "recipes," preserving perfect data fidelity across a network. In this model, intermediate routers forward packets based on symbolic headers without interpreting the payload. Separately, the invention of encoding map rotation provided a method for achieving forward secrecy on a point-to-point link by ensuring any single encoding map is used for only a limited time. However, there is a need to formally combine these two concepts to create a comprehensively secure, multi-hop networking protocol that is resistant to both payload analysis by intermediate nodes and retrospective decryption of captured traffic.

Summary of the Invention

The present invention is a secure network protocol that integrates the principles of parametric networking and encoding map rotation to create a system with end-to-end forward secrecy at the encoding layer.

In a network operating under this protocol, a sender and a receiver establish a communication session. They share a secret or method that allows them to synchronously generate a sequence of disposable encoding maps (Map A, Map B, Map C, etc.). Data is encapsulated in parametric packets, with payloads encoded using the currently active map (e.g., Map A). These packets are transmitted through one or more intermediate parametric routers. The routers operate solely on the packets' symbolic headers and are completely unaware of the payload's encoding scheme.

The protocol specifies a trigger condition for map rotation, such as a set amount of time elapsed or a number of packets sent. When the trigger condition is met, the sender and receiver both independently discard the current map (Map A) and switch to the next map in the sequence (Map B) for all subsequent communication in the session. This rotation renders captured historical traffic undecipherable, even if an attacker later compromises a

router and the map generation secret. The router's function is unchanged by the rotation, as it never interacts with the payload's encoding.

Brief Description of the Drawings

FIG. 1 is a system diagram illustrating two endpoints communicating through an intermediate parametric router, showing that the router is blind to the rotating encoding maps used for the payload.

Detailed Description of the Invention

Referring to **FIG. 1**, the operation of the secure protocol is shown. A Sender endpoint (101) and a Receiver endpoint (102) are engaged in a communication session across a network, with an intermediate Parametric Router (103) positioned in the data path.

The Sender (101) composes a parametric payload (104), which is a symbolic representation of data. This payload is encoded using a currently active encoding map, for example, 'Map A' (106). The encoded payload is then encapsulated in a parametric packet (105) which includes a symbolic header with routing information.

The packet (105) is transmitted to the Parametric Router (103). The router's Parametric Forwarding Engine (107) examines the packet's symbolic header only. Based on this header information and its internal routing table, it forwards the packet towards the Receiver (102) without inspecting, modifying, or attempting to decode the payload. The router has no knowledge of, or need for, 'Map A'.

The Receiver (102) receives the packet and, since it also has 'Map A', successfully decodes the payload. The protocol includes a pre-defined trigger condition (108), such as a timer or packet counter. When this trigger is met, both the Sender (101) and Receiver (102) are configured to discard 'Map A' and synchronously switch to the next map in their shared sequence, 'Map B'. Subsequent packets will be encoded and decoded using 'Map B', but the function of the Parametric Router (103) remains identical and unaffected. This ensures that a compromise of the map-generation secret at the time of 'Map B' cannot be used to decode traffic that was previously captured while 'Map A' was active.

Claims

What is claimed is:

1. A method for secure data communication in a multi-hop network, comprising the steps of:

    - at a sender device, encoding the payload of a data packet using a first disposable encoding map;
    - transmitting said data packet to an intermediate network router;
    - at the intermediate router, forwarding the data packet towards a receiver device based solely on header information, without reference to the encoding of the payload;
    - at the receiver device, decoding the payload of the data packet using the first disposable encoding map; and
    - upon the occurrence of a trigger condition, both the sender and receiver devices ceasing to use the first encoding map and commencing to use a second, different disposable encoding map for subsequent packets in the same communication session.

2. The method of claim 1, wherein the data packet is a parametric packet, the payload is a symbolic data representation, and the header information is a symbolic address.

3. The method of claim 1, wherein the trigger condition is selected from the group consisting of: a predetermined amount of time having elapsed, and a predetermined number of data packets having been transmitted.

4. The method of claim 1, further comprising the step of the sender and receiver devices securely discarding the first encoding map after commencing use of the second encoding map.

5. A secure networking system, comprising:

    - a sender device configured to encode a packet payload with one of a sequence of disposable encoding maps;
    - a receiver device configured to decode the packet payload with the same disposable encoding map;

- wherein both the sender and receiver are configured to synchronously switch from a current encoding map to a next encoding map in the sequence upon a trigger condition being met; and
- an intermediate router configured to forward packets from the sender to the receiver based on packet header information, wherein the router has no knowledge of the encoding maps used by the sender and receiver.

6. The system of claim 5, wherein the intermediate router is a parametric router that operates on symbolic header addresses.

7. The system of claim 5, wherein the sequence of disposable encoding maps is generated from a shared secret between the sender and receiver.

Drawings

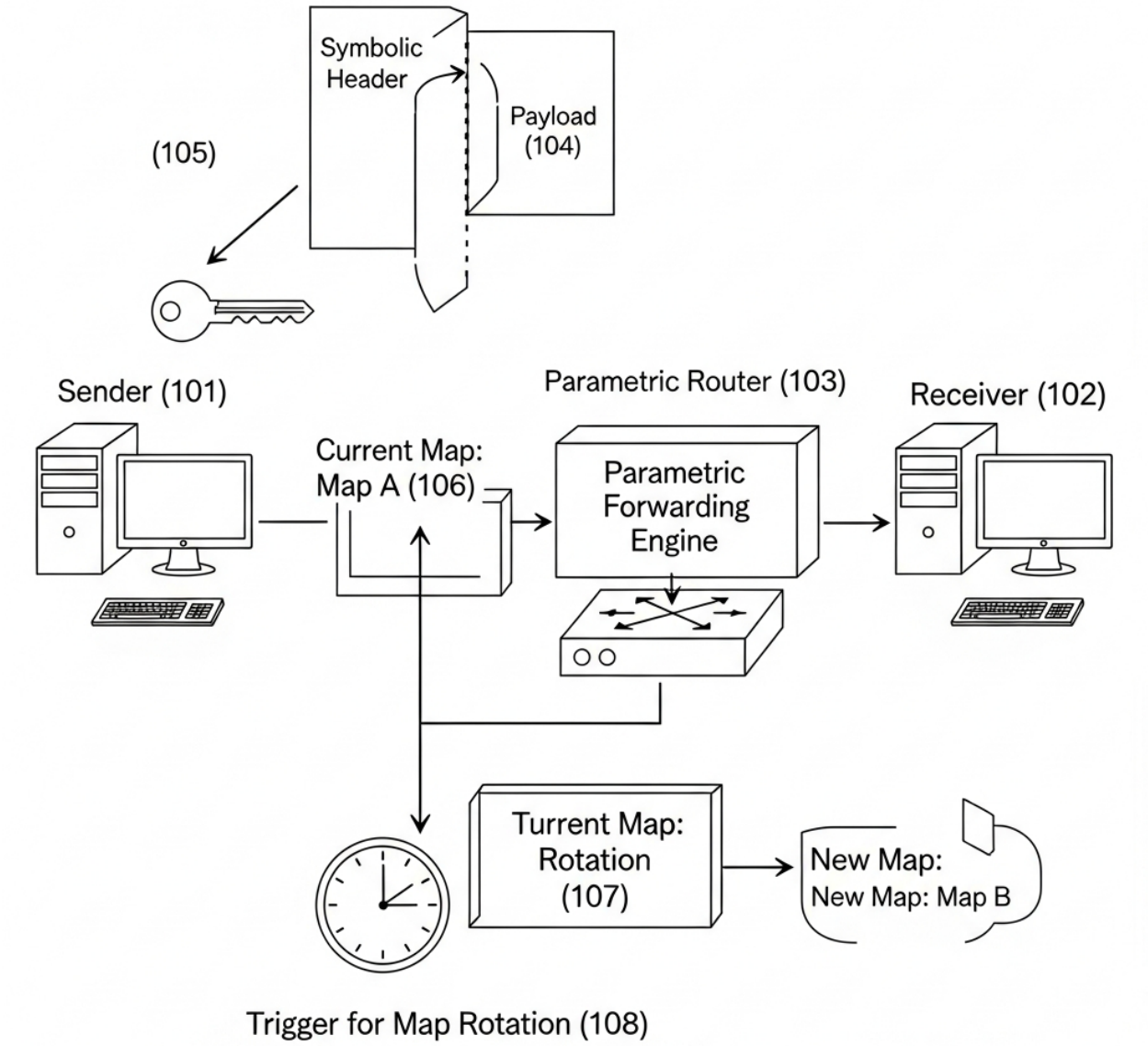


Figure 1: System diagram showing endpoints communicating through a router blind to the rotating payload encoding maps.